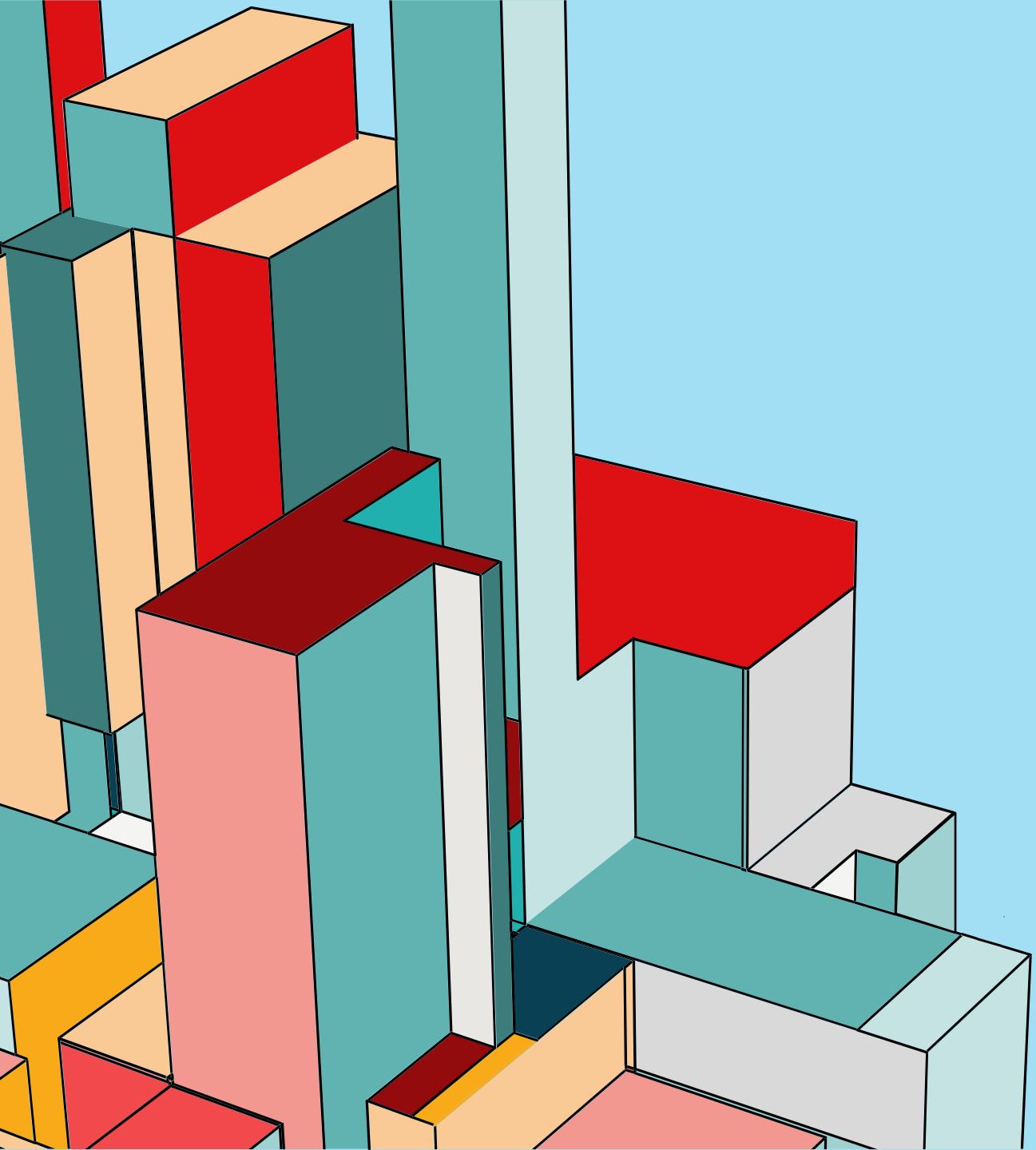


APPLIED DATA SCIENCE PROJECT

GEO-EMBEDDINGS AS PRIOR FOR BURNED AREA DELINEATION

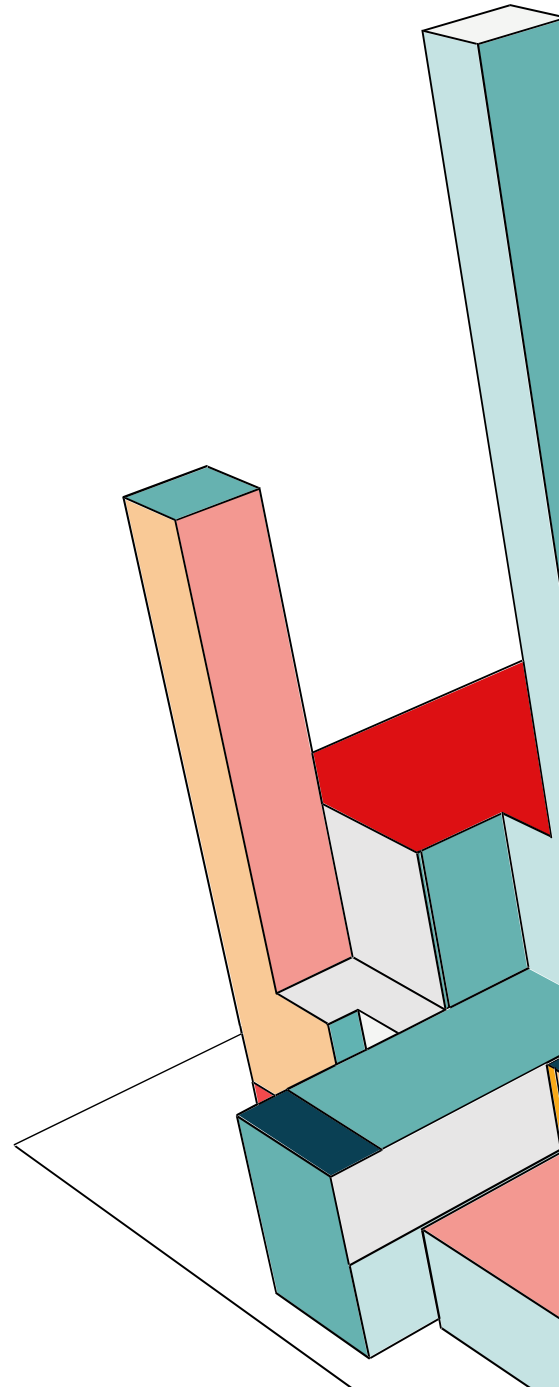
Politecnico di Torino

FONDAZIONE links PASSION FOR INNOVATION

ellis European Laboratory for Learning and Intelligent Systems

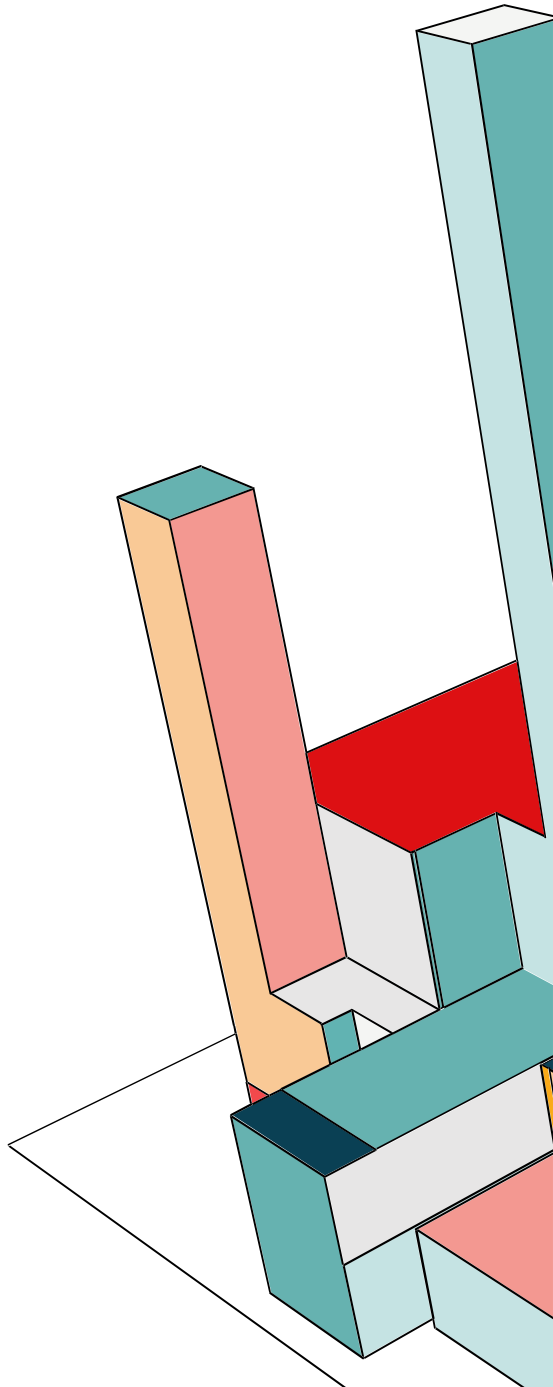
VALUE-DRIVEN PROJECT

- **Geo-embeddings** are compact per-pixel representations produced by EO foundation models (e.g. AlphaEarth Satellite Embeddings, yearly, 10 m), summarizing land cover, vegetation and terrain
- **Burned area delineation** objective is to segment the area affected by a wildfire from post-fire satellite images
- The aim of this project is to assess whether geo-embeddings are a useful **prior** for fire estimation, even when produced **before the fire** and thus without its spectral signature



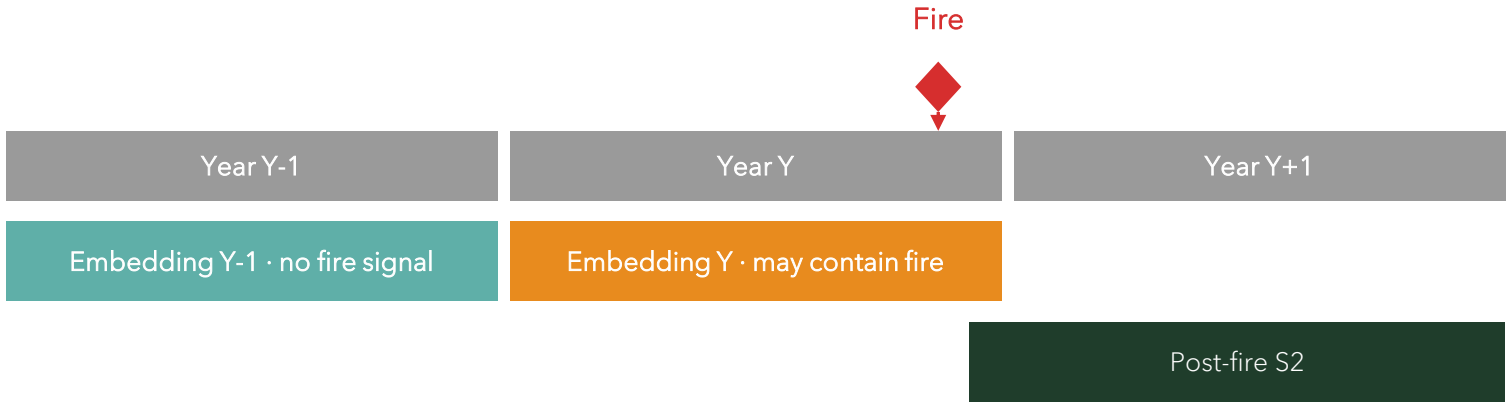
DATA

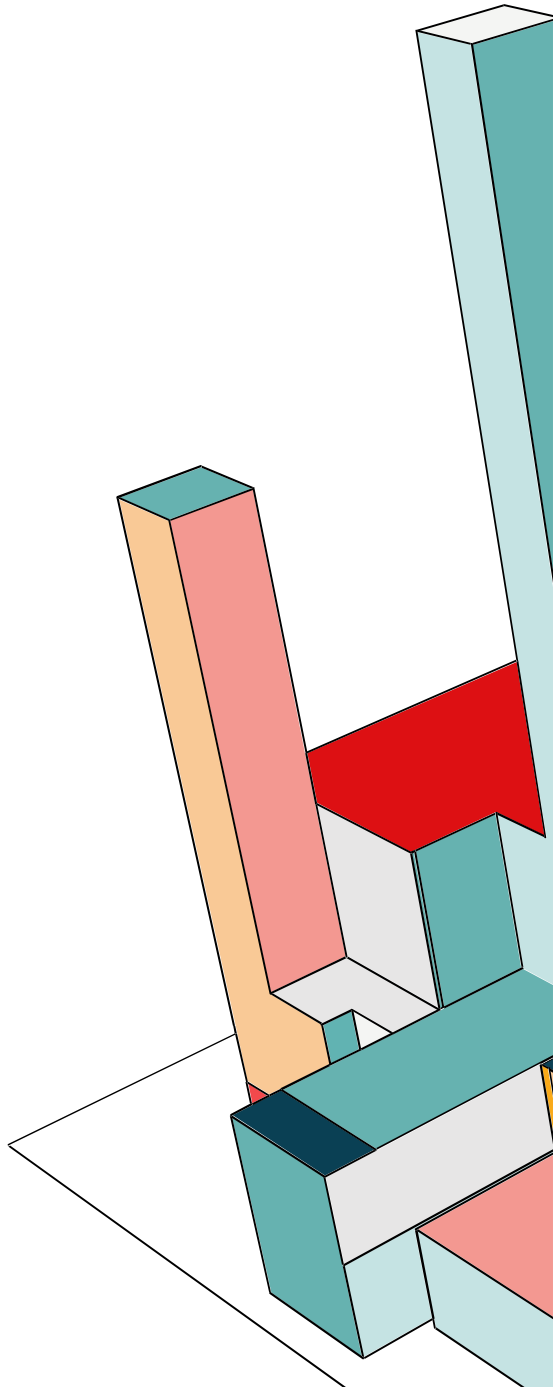
- We will use a catalog of fires happened in **Europe**
- For each fire event, the following data are available
    - *Sentinel-2 L2A images (pre- and post-fire) with cloudmask*
    - *Dynamic World Land Cover*
    - *Annual geo-embeddings (e.g. AlphaEarth, available from 2017)*
    - *Burned area delineation mask (Ground truth)*
    - *Validity mask*



TASK

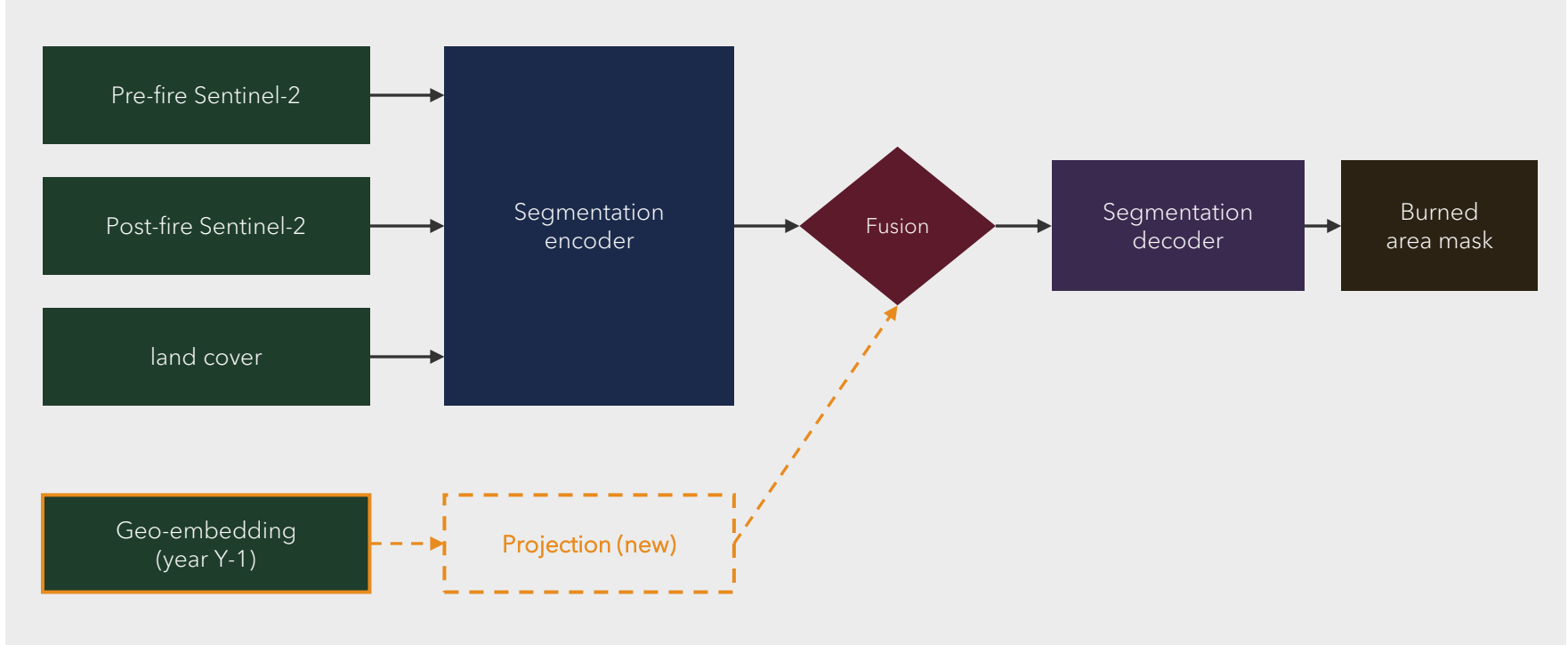
- Compare burned area delineation with different **embedding priors**
  - No embeddings (baseline)
  - Embeddings of the year **before** the fire (no fire signal)
  - Embeddings of the **same** year (may contain the fire)
- Is the prior still useful when it **doesn't see the fire**?

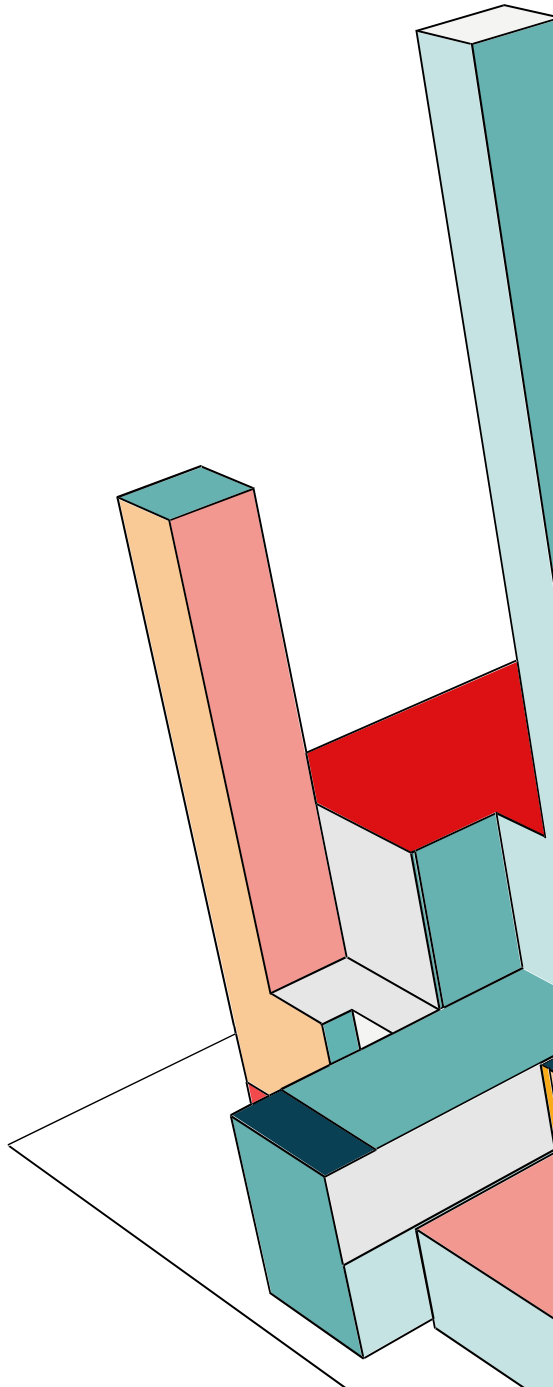




TASK

- Integrate the embeddings as prior into a segmentation model
    - e.g. early fusion, late fusion, embedding-conditioned decoder. **Ideas welcome!**





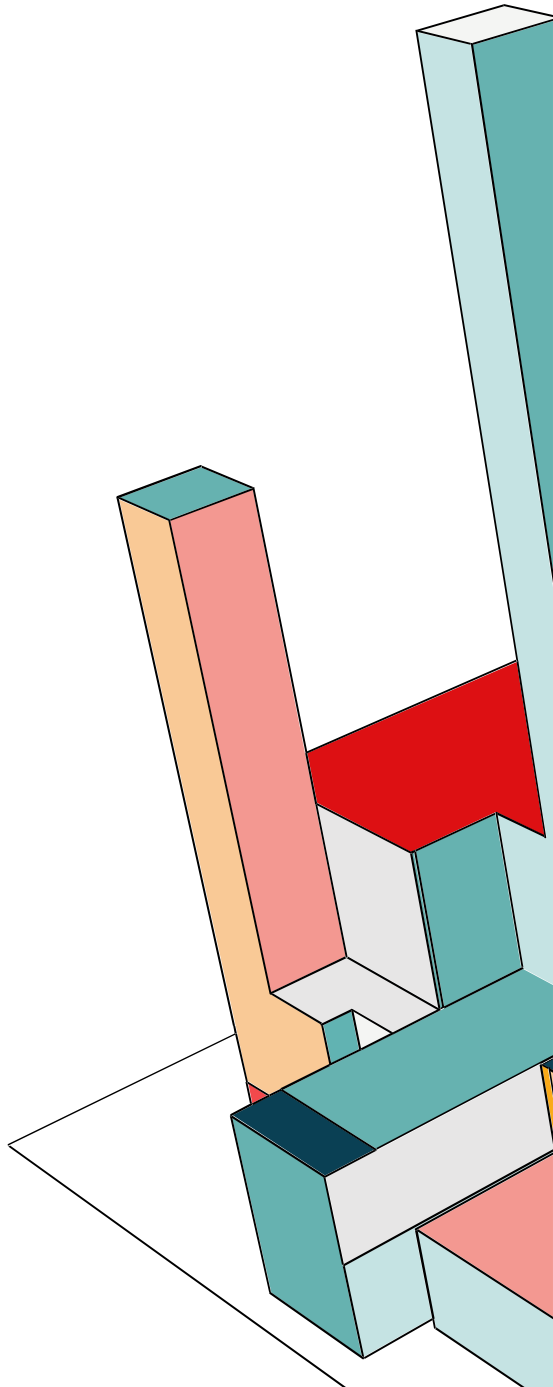
LIGHT MENTORING

Mentors

- Luca Barco *luca.barco@linksfoundation.com*

Weekly 30 min calls with students for the whole duration of the semester

Feel free to reach out via **Slack** or **email** at any time for any questions or doubts



POLICY

- Both project descriptions and implementations will be part of a repository group published on GitHub

- The repositories will be public

